

Python Cheat Sheet 1:

Print():

- `print("Hello World!")`
- A way to tell the user something in the terminal.
- A common debugging tool.

Input():

- `input("What is your name? ")`
- A way to ask the user for input/data.

Basic Data Types:

- String: Text. Surrounded by quotation marks. "Hello World!"
- Integer: A whole number. 6
- Float: A Decimal number. 3.6666
- Boolean: True/False

Variable:

- A space in the computer's memory
- `Score = 5`

Text Escape sequences(Backslash):

- `\n` : creates a line return in a string of text
- `\t` :creates a tab style indent in a string of text
- `\\` : allows a backslash to appear in a string of text
- `\"` : allows a speech mark to be used in a string of text
- `print("I say "High", you say "Low". You say "Why?" and I say "I don't know". Oh no.)`

Math operators:

Operator	Name:	Example:	Answer:
<code>*</code>	Multiply	<code>2*3</code>	6
<code>/(forwardslash)</code>	Divide(Normal)	<code>20/8</code>	2.5
<code>//</code>	Divide(integer)	<code>20/8</code>	2
<code>%</code>	Modulus	<code>20%8</code>	4
<code>+</code>	Add	<code>2+3</code>	5
<code>-</code>	Subtract	<code>7-3</code>	4

Logical Operators:

Operator	Meaning
<code>==</code>	Equal to
<code>!=</code>	Not equal to
<code>></code>	Greater than
<code><</code>	Less than

>=	Greater than or equal to
<=	Less than or equal to

While Loop:

- A piece of code that runs over and over again while a statement is true.
- greeting = "Hi"
- count = 0
- while count<5:
- print(greeting)
- count = count+1

Puzzle 1:

Write some code in IDLE so that the computer counts up to 20 in twos.

Puzzle 2:

Write some code so that the computer outputs the 5 times table like this.

1x5=5

2x5=10

3x5=15

Hint: You will need a counter variable which you could call number. Then you should find out how to write one line, and then make your loop do it 10 times.

Puzzle 3:

See if you can re-write the following code in three different ways so that each program still produces output which counts to a hundred.

```
>>> number=1
```

```
>>> while number < 101: print(number) number = number+1
```

In your new code, you are not allowed to use the less than operator <. Instead you should use one of these comparative operators in each program: <= > !=